Week 1 — Introduction

OpenMV

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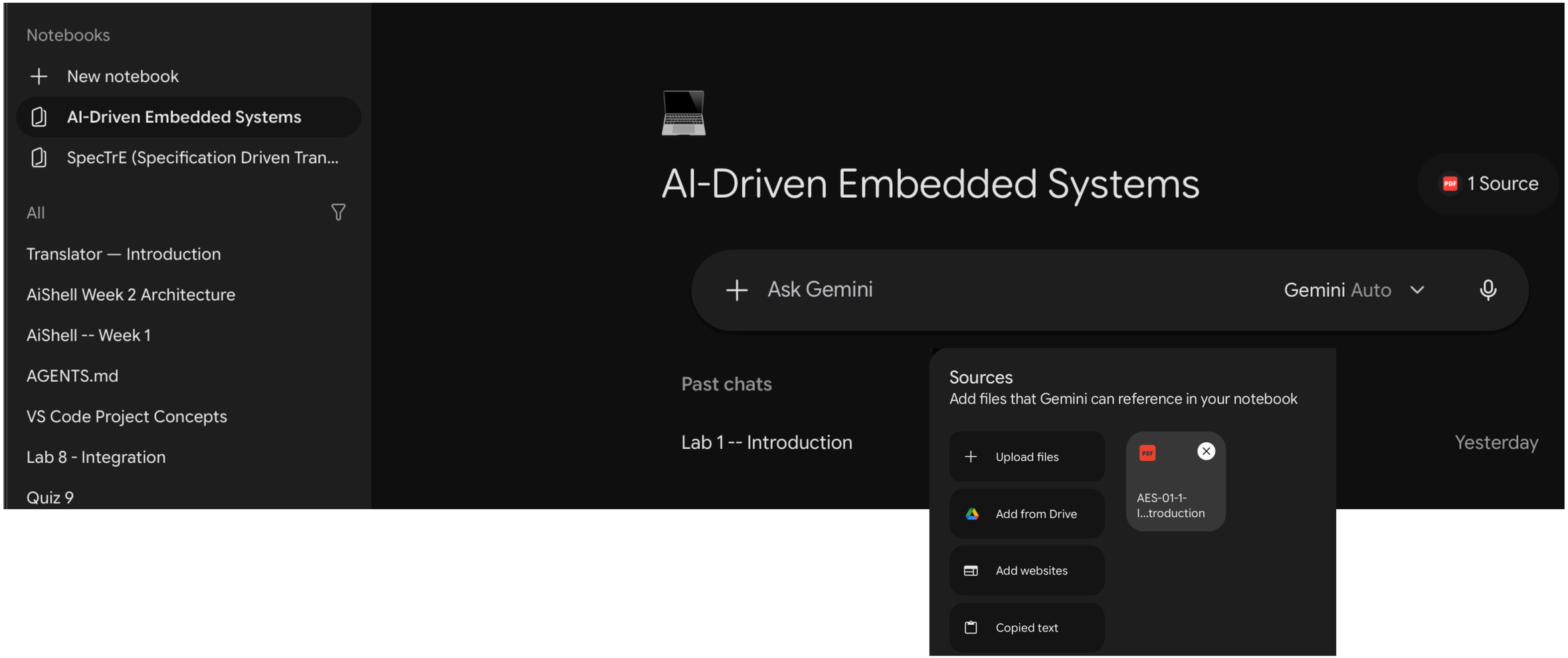
Gemini Note — Notebooks

- Help organize projects; support sources & chats

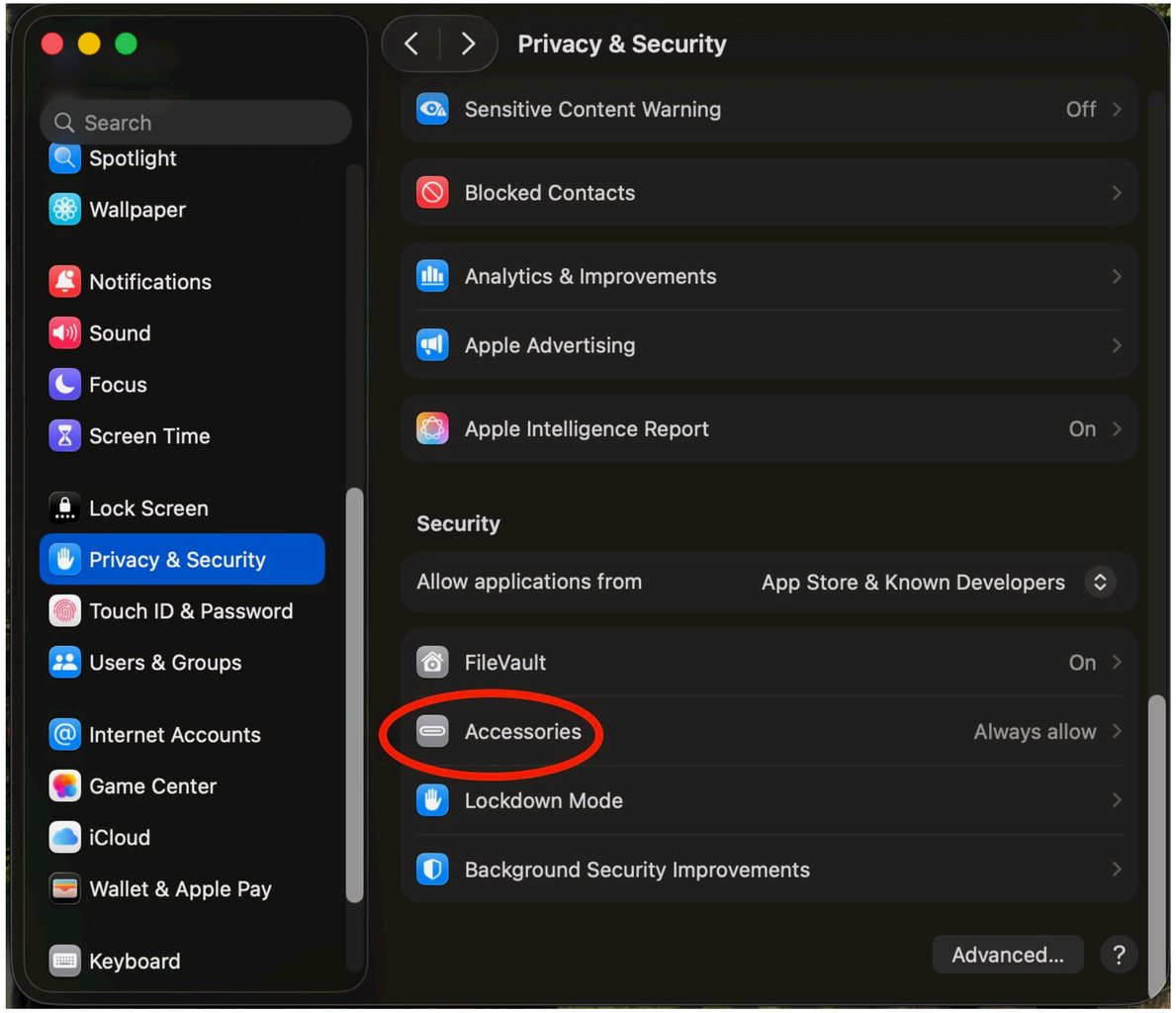


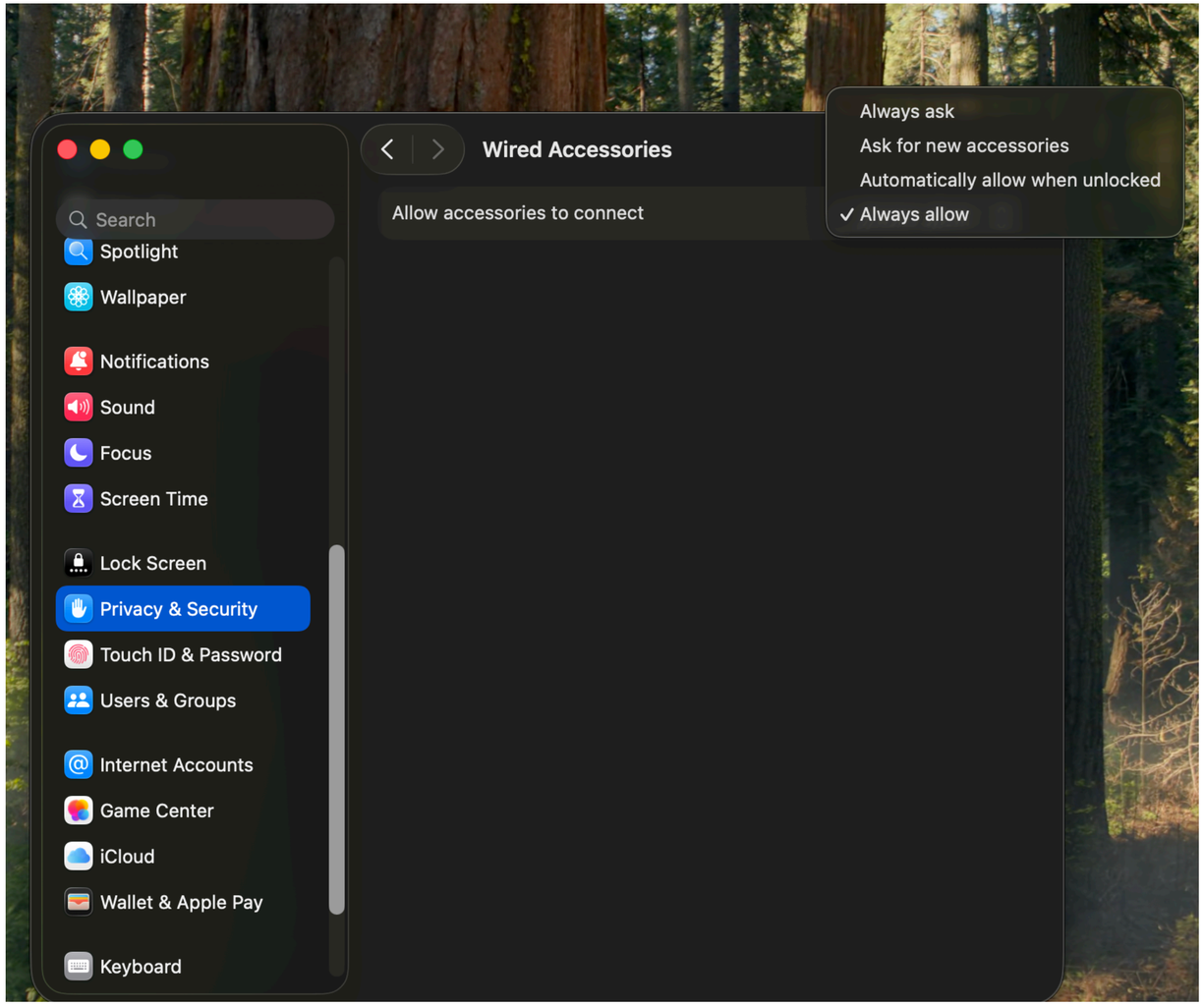
OpenMV Cam N6

Connection:

USB-C to USB-C Cable

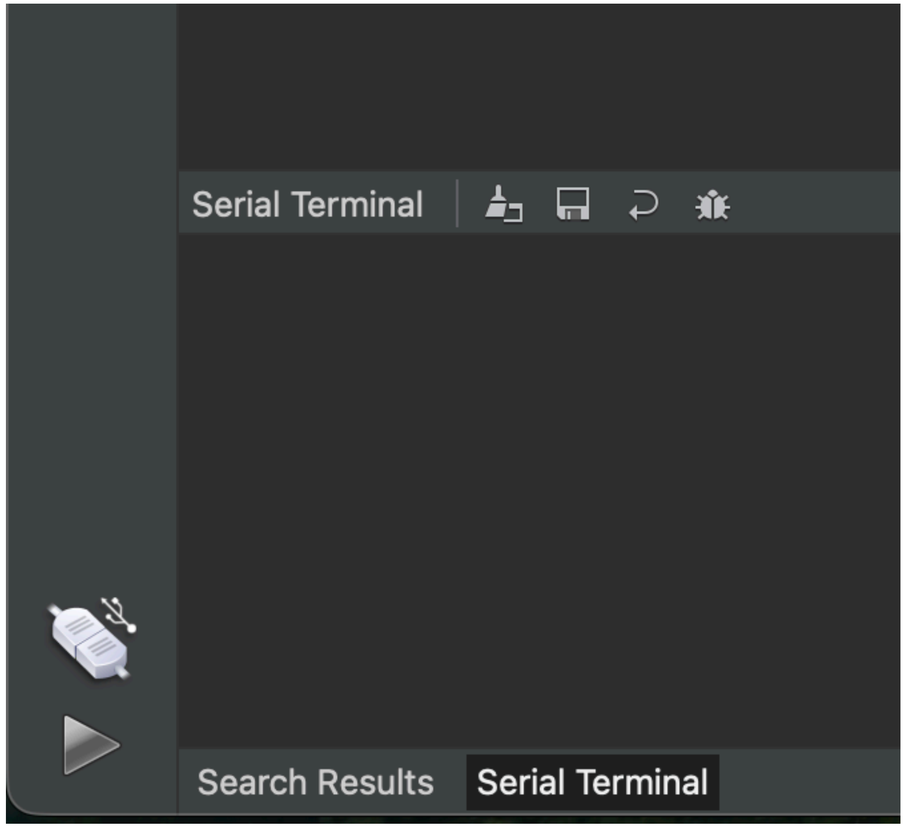
MacOS Connection Settings



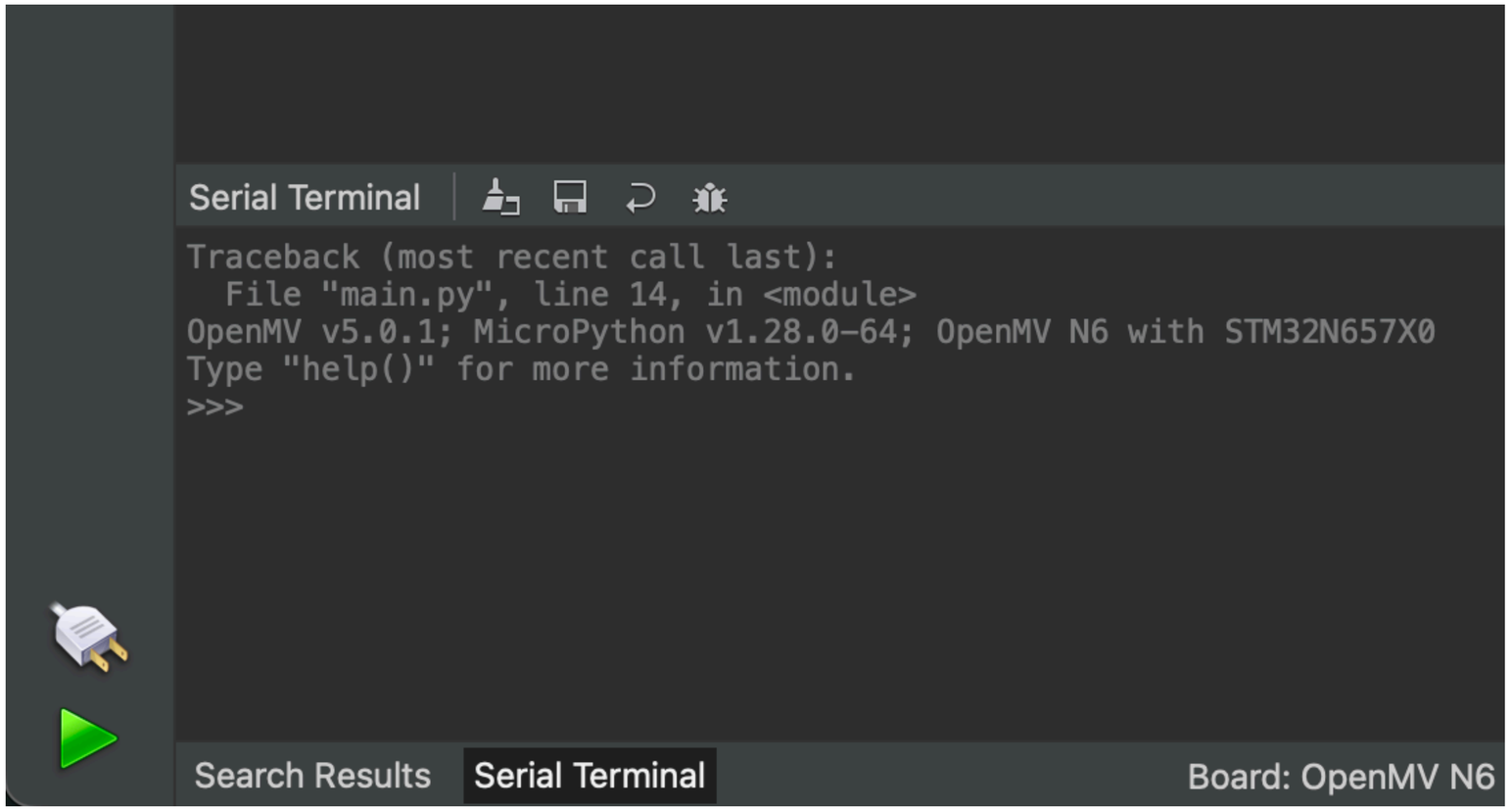


Connecting N6 Cam to OpenMV IDE

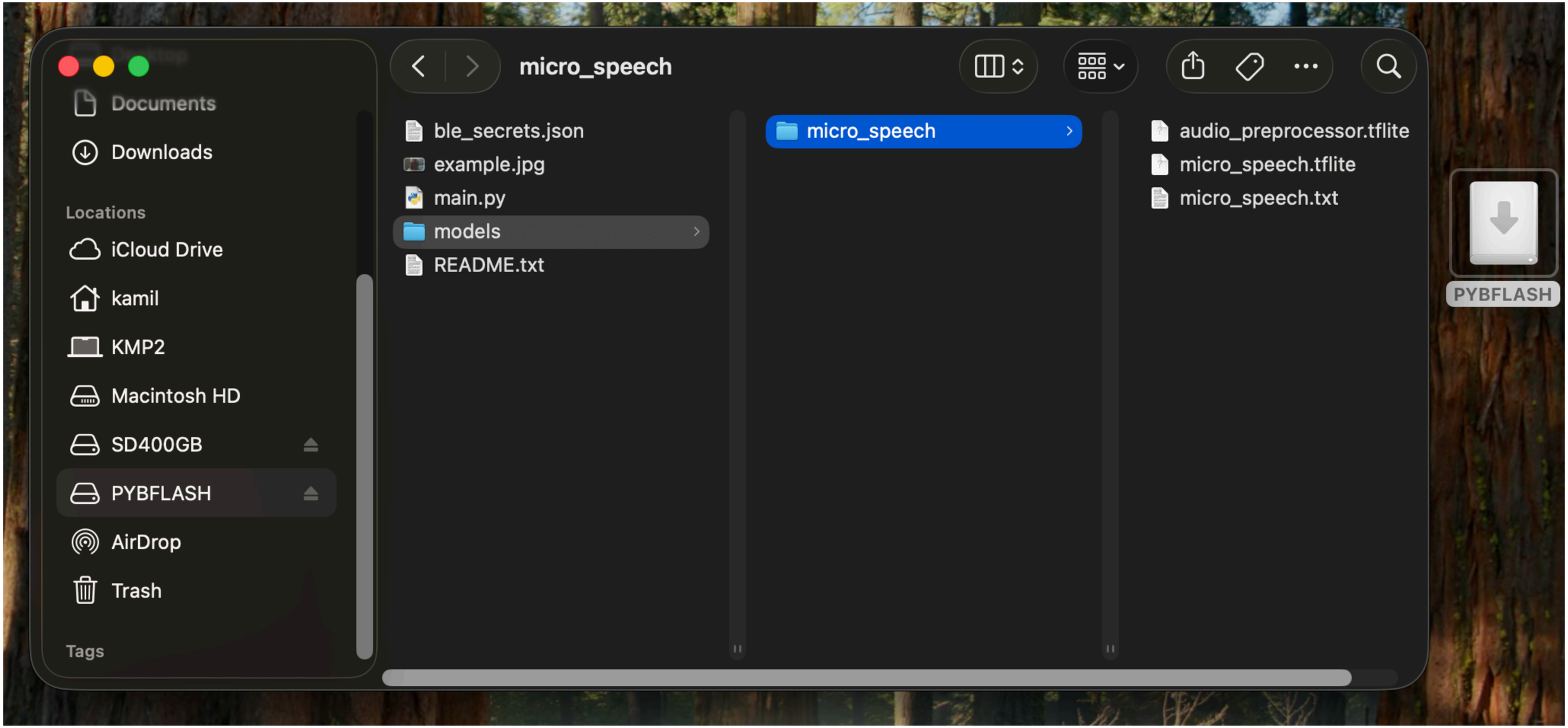
Unconnected



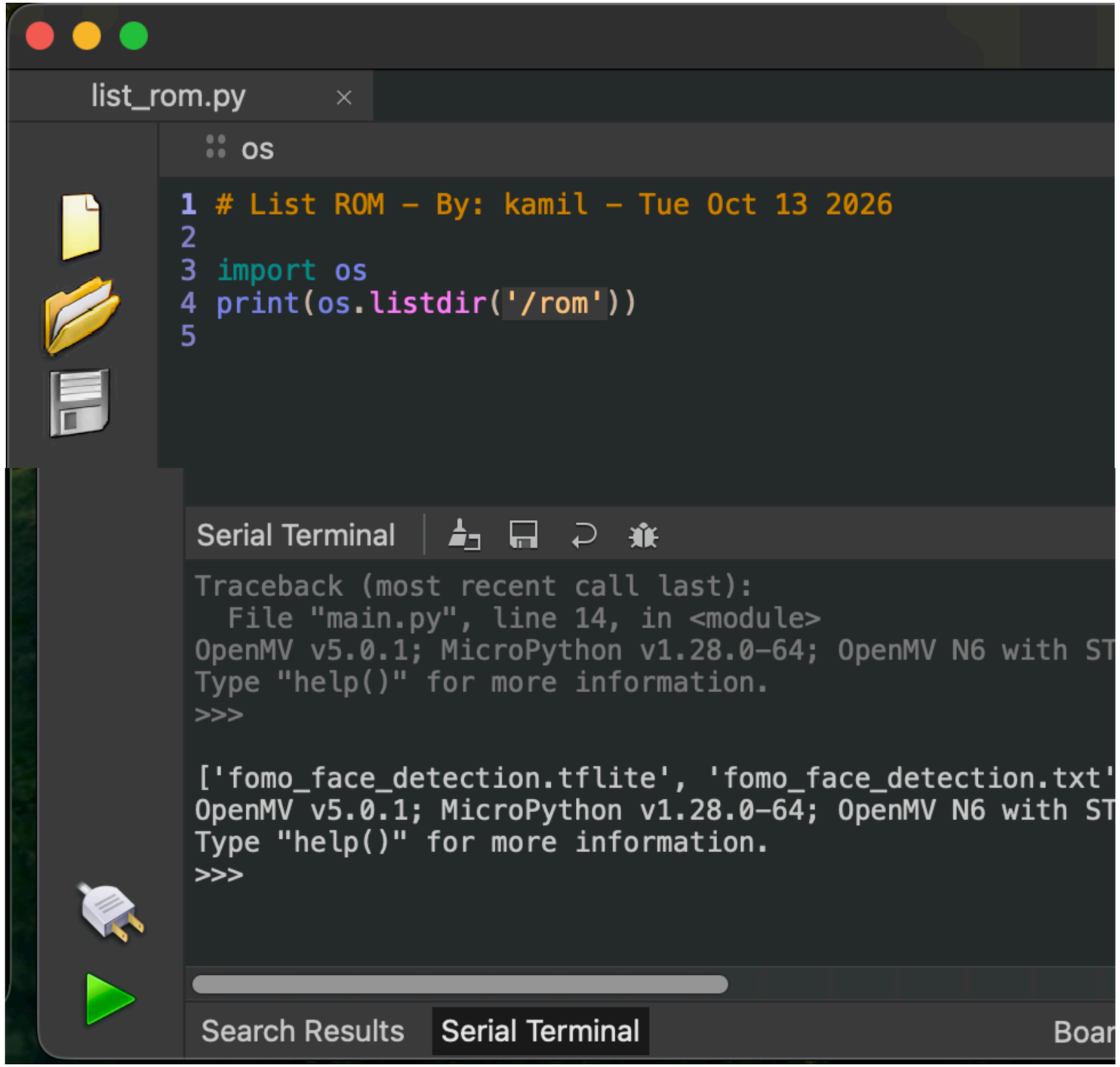
Connected



SD Card mounted as PYBFLASH



ROM Listing


list_rom.py
os
1 # List ROM - By: kamil - Tue Oct 13 2026
2
3 import os
4 print(os.listdir('/rom'))
5
Serial Terminal
Traceback (most recent call last):
File "main.py", line 14, in <module>
OpenMV v5.0.1; MicroPython v1.28.0-64; OpenMV N6 with ST
Type "help()" for more information.
>>>
['fomo_face_detection.tflite', 'fomo_face_detection.txt'
OpenMV v5.0.1; MicroPython v1.28.0-64; OpenMV N6 with ST
Type "help()" for more information.
>>>
Search Results
Serial Terminal
Boar

tf_object_detection_1.py - OpenMV IDE

tf_object_detection_1.py*

csi — Line: 26, Col: 9

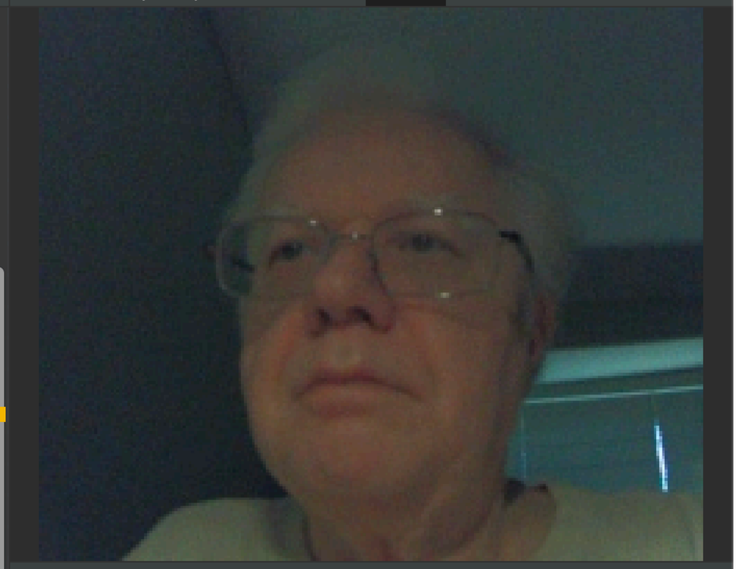
```
17 csi0.pixformat(csi.RGB565)  # Set pixel format to RGB565 (or GRAYSCALE)
18 csi0.framesize(csi.QVGA)  # Set frame size to QVGA (320x240)
19 csi0.window((240, 240))  # Set 240x240 window.
20 csi0.snapshot(time=2000)  # Let the camera adjust.
21
22 # Load built-in FOMO face detection model
23 model = ml.Model("/rom/fomo_face_detection.tflite", postprocess=Fomo(threshold=0.7))
24 print(model)
25
26 4444444$# Alternatively, models can be loaded from the filesystem storage.   invalid syntax   E261 at least two s…
27 # model = ml.Model('<object_detection_modelwork>.tflite', load_to_fb=True)
28 # labels = [line.rstrip('\n') for line in open("labels.txt")]
29
30 colors = [  # Add more colors if you are detecting more than 7 types of classes at once.
31     (255, 0, 0),
32     (0, 255, 0),
33     (255, 255, 0),
34     (0, 0, 255),
35     (255, 0, 255),
36     (0, 255, 255),
37     (255, 255, 255),
38 ]
39
40 clock = time.clock()
41 while True:
42     clock.tick()
43     img = csi0.snapshot()
44
45     for i, detection_list in enumerate(model.predict([img])):
46         if i == 0:
47             continue  # background class
48         if len(detection_list) == 0:
49             continue  # no detections for this class?
50
51         print("********** %s **********" % model.labels[i])
52         for (x, y, w, h), score in detection_list:
53             center_x = math.floor(x + (w / 2))
54             center_y = math.floor(y + (h / 2))
55             print(f"x {center_x}\ty {center_y}\tscore {score}")
56             img.draw_circle((center_x, center_y, 12), color=colors[i])
57
58     # print(clock.fps(), "fps", end="\n")
59
```

Serial Terminal

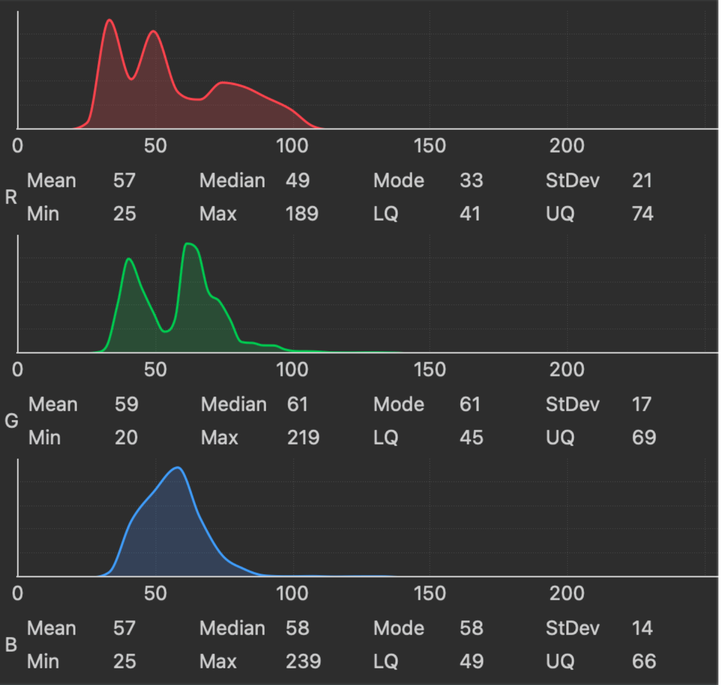
```
OpenMV v5.0.1; MicroPython v1.28.0-64; OpenMV N6 with STM32N657X0
Type "help()" for more information.
>>>


{ model_size: 64064, model_addr: 0x70800040, ram_size: 29440, ram_addr: 0x3410a3e0, input_shape: ((1, 96, 96, 3),), input_s
********** face **********
x 68    y 140   score 0.7382814
********** face **********
x 68    y 123   score 0.7265626
********** face **********
x 68    y 140   score 0.7265626
```

Frame Buffer (JPEG) - Re… | Record | Zoom | On | JPG Mode



Histogram — RGB Color Space



Search Results | Serial Terminal | Board: OpenMV N6 | Sensor: PAG7936 | Firmware Version: 5.0.1 - [latest] | Serial Port: cu.usbmodem2101 | Drive: /Volumes/PYBFLASH | FPS: 117.9 Cam - 76.9 IDE